FarmCD: Agricultural Information-Guided Gated Network for Farmland Change Detection From Remote Sensing Images

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***Abstract*—In recent years, farmland change detection (CD) from remote sensing (RS) images has emerged as an important research topic. Due to the complexity of farmland dynamics, the existing CD methods often perform poorly when directly applied to farmland scenarios. To address these challenges, FarmCD is proposed as a farmland-specific CD framework that selectively focuses on farmland regions while suppressing irrelevant categories, such as buildings. FarmCD integrates several specialized modules to extract and refine farmland-related change information. The farmland feature aware module (FFAM) extracts farmland features from bi-temporal images through two mutually interactive branches, while the differential fusion module (DFM) captures temporal variations and structural differences. To mitigate interference from nonfarmland categories, the farmland soft–hard gate module (FSGM) adaptively suppresses nonfarmland changes. In addition, the multiscale farmland refinement module (MFRM) progressively enhances farmland boundary details, reducing misdetections caused by irregular farmland shapes. By integrating weakly supervised farmland mask generation and a multistage training strategy, the network dynamically filters and enhances farmland-relevant features throughout the CD process. Experimental results on the public cropland CD (CLCD) and cropland semantic CD (CropSCD) datasets demonstrate that FarmCD achieves competitive performance, with significant improvements over seven state-of-the-art (SOTA) models. The implementation of FarmCD will be publicly available in the future at https://github.com/lixinghua5540/FarmCD**

***Index Terms*—Attention mechanism, farmland change detection (CD), multistage training strategy, remote sensing (RS), weakly supervised learning.**

I. Introduction

FARMLAND is the cornerstone of global food production, directly providing the majority of the world's food and linking agricultural resources closely to food security [1]. However, rapid economic development, together with natural degradation and anthropogenic pressures, such as urbanization and industrial expansion, has led to the continuous decline of farmland resources [2]. In this context, accurate and timely farmland change detection (CD) has become crucial for land use monitoring, crop dynamics assessment, and sustainable agricultural management [3]. By identifying farmland expansion, degradation, and structural transformation, it provides critical support for understanding agricultural landscape dynamics and evaluating the effectiveness of agricultural policies, especially under the dual pressures of rising food demand and environmental challenges.

Traditional CD techniques in remote sensing (RS) are generally divided into three categories: algebra-based [4], [5], transformation-based [6], [7], and postclassification methods [8]. While these methods are simple and efficient, they often rely on empirical thresholds or handcrafted features, making them subjective and less robust, especially in high-resolution imagery with complex object distributions. Machine learning approaches, such as support vector machine [9] and random forest [10], improved performance to some extent but still struggled with spatial context modeling.

In recent years, deep learning (DL) has significantly advanced remote sensing CD (RSCD), overcoming many limitations of traditional and machine learning-based methods [11]. By automatically extracting hierarchical features, DL enables end-to-end frameworks and achieves superior performance, particularly on high-resolution imagery [12]. Convolutional neural networks (CNNs) dominate RSCD tasks due to their abilities to capture both low-level spatial details and high-level semantic information [13]. Early DL-based methods typically adopt early fusion, concatenating co-registered bi-temporal images along channels for a single-branch network [14]. However, early fusion often fails to explicitly model temporal differences. Later fusion strategies address this limitation, using Siamese or dual-branch architectures, where each branch extracts features from one temporal image and differences are fused in deeper layers [15], [16]. Later fusion networks consistently outperform early fusion in capturing discriminative change patterns, as demonstrated by FC-Siam-conc and FC-Siam-diff [17]. To further enhance feature extraction, attention mechanisms [18], convolutional block attention modules, and deep supervision [19] have been integrated to focus on salient regions and improve gradient flow. Dense skip connections, as in SNUNet [20] and UNet++ [21], combine multilevel features to improve localization, while pyramid structures [22] and dilated convolutions [23] enlarge receptive fields efficiently.

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Simultaneously, the explosive growth of RS data has enabled the acquisition of multimodal RS imagery within the same region. Researchers employ feature alignment techniques to fuse multimodal imagery, such as SAR imagery [24] and near-infrared data [25], thereby enhancing the image feature extraction and data fusion capabilities of DL models.

Transformer-based models, such as BIT [26], [27], leverage self-attention to capture long-range spatiotemporal dependencies. Despite these advances, challenges remain: many models underutilize temporal differences and multiscale fusion often relies on fixed designs without fully exploring interscale interactions. Recently, as certain CD tasks (e.g., building CD) reached high accuracy, research focus has shifted to efficiency. Superpixel-based global energy optimization [28], vision foundation model-based networks based on FastSAM [29], [30], and radiometric normalization for generalization [31] illustrate approaches to accelerate inference and maintain accuracy with relatively simple architectures.

Although RSCD methods based on DL have become increasingly mature, many challenges remain for farmland CD. First, conventional CD networks, such as FC-EF [17] and A2Net [32], tend to perform poorly on farmland-specific datasets. For instance, their recall rates often remain around 40%, and the $F1$-scores are typically only about 50%, indicating the difficulty in distinguishing subtle agricultural changes from other types of land cover variations. Accurately and efficiently detecting farmland changes remains a pressing challenge. To address such complex data challenges, some researchers have explored the fusion of optical and SAR imagery [33], while others have directly employed uncrewed aerial vehicle [34] imagery to perform fine-grained semantic CD in agricultural scenarios. Meanwhile, in response to the complexity of agricultural change types and the scarcity of annotated data, several recent studies have proposed customized solutions specifically tailored for farmland CD. For instance, Liu et al. [35] introduced MeGNet, a memory-guided network designed for semantic farmland CD. The core of MeGNet lies in its class-aware memory module, which effectively captures and differentiates semantic representations of various change categories. This approach helps mitigate the difficulties arising from diverse farmland change patterns and limited labeled samples. Similarly, Liu et al. [36] proposed a CNN–transformer hybrid network featuring multiscale context aggregation, specifically optimized for farmland CD tasks. Their model leverages CNNs to extract hierarchical spatial features and incorporates a transformer-based multiscale contextual attention module to encode and fuse contextual information across scales. A multibranch CNN-based classification head is then employed under a deep supervision strategy to produce accurate change maps. Recent research on farmland CD has explored new directions. Xing et al. [37] proposed a network architecture focused on farmland boundaries to address the low accuracy of detecting irregular farmland changes. Lv et al. [38] introduced a dual-backbone network to balance the tasks of cultivable land feature extraction and CD. These efforts highlight the trend toward designing specialized architectures that integrate semantic modeling and contextual attention to address the unique challenges of farmland CD.

Although notable progress has been made in farmland CD research, significant challenges still remain. Factors, such as crop rotation, spectral variations across different crop growth stages, complex crop types, and the fragmented distribution of arable land, significantly reduce detection accuracy compared to CD in other land categories [39].

To address these challenges, this study introduces a CD network specifically designed for farmland scenarios, termed FarmCD. The network is tailored to focus solely on farmland-related changes, intentionally excluding irrelevant categories, such as buildings or other nonagricultural transformations. This selective focus is particularly valuable for tasks, such as monitoring the nonagriculturalization and nongrain land encroachment of arable land. FarmCD integrates several farmland-specific modules into a unified framework, including farmland feature aware module (FFAM), differential fusion module (DFM), farmland soft–hard gate module (FSGM), and multiscale farmland refinement module (MFRM). The main contributions are as follows.

1) A parallel multitask framework is proposed, in which weakly supervised farmland detection is used to guide the farmland CD process. Through joint training, a strong correlation is established between the CD task and the target region (i.e., farmland), enabling the network to obtain more accurate change information.
2) FFAM and DFM are introduced, where FFAM is designed for farmland feature extraction and DFM enhances discriminative change cues, thereby improving the extraction of agricultural region-related change features.
3) FSGM is designed to dynamically fuse farmland masks with change features, performing region-aware feature modulation and improving the network's ability to focus on and respond to agricultural changes.

The remainder of this article is organized as follows. Section II presents the proposed method. Section III introduces the comparative experiments and analyzes the results. Section IV presents several key findings. Finally, the conclusions are drawn in Section V.

II. Methodology

This section provides an overview of the architecture of FarmCD. To address issues, such as false positives (FPs) and false negatives (FNs) in farmland CD, this article designs a dual-stream structure. While extracting differential fusion features from bi-temporal images, a farmland feature extraction module is introduced to guide the differential fusion features in various ways based on the extracted farmland information. Finally, a refinement module is employed to polish the detection results, achieving high-accuracy detection outcomes.

A. Architecture Overview

This article focuses specifically on farmland, designing a farmland feature extraction module that guides the network to pay attention to farmland changes through the extracted features. Ultimately, it precisely captures farmland changes

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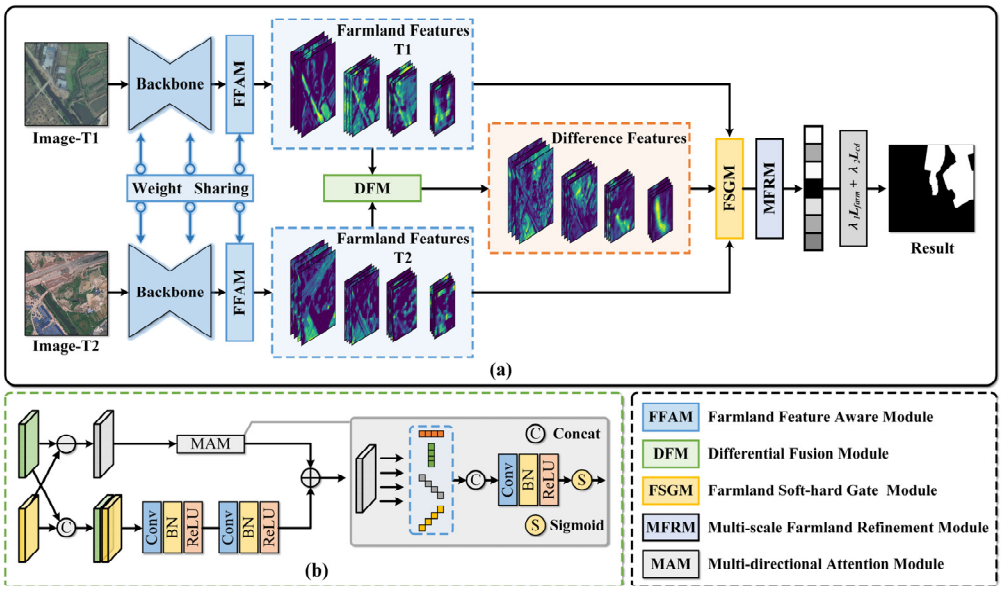


Fig. 1. (a) Main architecture of FarmCD. (b) DFM.

while avoiding interference from changes in other land types. The overall architecture of FarmCD is illustrated in Fig. 1. Aiming to enhance the perception of changes within farmland regions, FarmCD introduces an explicit farmland semantic guidance mechanism to achieve more robust CD performance under complex backgrounds.

In the feature extraction stage, FarmCD builds a dual-branch structure to separately extract farmland features from the bi-temporal RS images. The farmland feature extractor generates multiscale feature representations across four stages, with spatial resolutions of 64×64, 32×32, 16×16, and 8×8, and corresponding channel dimensions of 512, 320, 128, and 64, respectively. It is particularly important to note that although the farmland feature extractors for the T1 and T2 images share the same architectural design, their weights are not shared. This design choice enhances the network's ability to learn more discriminative feature representations. The extracted multilevel farmland features are then fed into a differential change feature extraction module to explicitly model the temporal differences, generating an initial representation of change. To further enhance the discriminative ability of the change features, the farmland features are subsequently passed through a custom-designed gating mechanism, which guides the learning of differential features by directing the model's attention toward true changes in farmland regions while suppressing interference from nonfarmland areas. Next, the fused features are processed by MFRM to enhance boundary details and improve the spatial consistency of the CD results. Finally, the outputs from all scales are upsampled and integrated to generate the final change map.

B. Feature Extraction Module

In CD tasks, the feature extraction module plays a pivotal role, as it directly determines the network's ability to perceive and discriminate change regions. It also significantly impacts the model's convergence speed, training stability, and generalization performance. In the specific context of farmland CD, this task becomes particularly challenging due to the complex composition of land cover in RS imagery, as well as the fragmented and heterogeneous distribution of farmlands.

First, different types of farmlands (e.g., paddy fields, dry land, and facility-based agriculture) exhibit significant variations in spectral, textural, and structural characteristics, resulting in low intraclass similarity. Second, due to the fragmented nature of arable land and variations in plot scale, even farmland of the same category may display diverse appearances in RS images. Additionally, spectral similarities between farmland and other land types (such as forests or algae-covered ponds) can lead to high interclass similarity, increasing the risk of false detections and omissions in the CD process.

Against this backdrop, effectively extracting discriminative and multiscale land cover features under limited data conditions is essential to improve model performance. To this end, a multiscale farmland-aware feature extractor is proposed in this study to enhance the network's capacity to perceive complex land features and distinguish meaningful changes. Specifically, the bi-temporal RS images, T1 and T2, are first fed into a dual-branch feature extraction module, which consists of two key subcomponents: FFAM and DFM.

FFAM takes both temporal images as an input and focuses on extracting stable semantic representations of farmland areas. These features are later used to guide the CD process and enhance the saliency of change regions. Meanwhile, DFM concentrates on capturing dynamic changes between the two timestamps, such as transitions in land cover type and structural alterations in farmland.

For the change feature extraction process, images T1 and T2 are passed through a shared-weight pyramid backbone network. This backbone employs an improved version of the pyramid vision transformer (PVT_V2), pretrained on ImageNet [40]. Let $F^i \in R^{H \times W \times C} (i \in \{1, 2, 3, 4\})$ represent the four levels of features extracted from shallow to deep layers;

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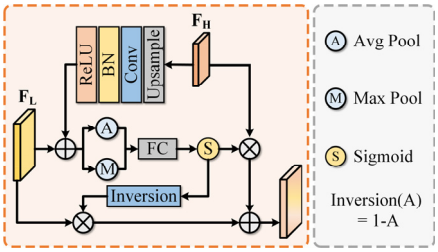


Fig. 2. FFAM.

these features are obtained from both T1 and T2, covering a range of receptive fields and semantic resolutions.

Each pair of features at the same scale is concatenated to form a set of multiscale feature pairs. These are then enriched with farmland-aware features extracted by FFAM and subsequently passed to the proposed DFM. DFM utilizes a multidirectional attention mechanism to model intertemporal interactions, effectively capturing spatiotemporal patterns of change and constructing robust differential representations. Experiments show that multidirectional attention aligns more closely with the diagonal texture patterns commonly observed in farmland imagery, thus better meeting the needs of farmland CD.

Through this dual-path feature extraction strategy, the network not only captures discriminative change features effectively but also leverages semantic farmland guidance to preliminarily screen and assess change regions. This provides a strong feature foundation for subsequent modules, including guided fusion and boundary refinement.

1) Farmland Feature Aware Module: To enhance the model's sensitivity to semantically relevant farmland regions, FFAM is introduced in Fig. 2. This module performs adaptive fusion between high-level semantic features and low-level spatial features. Specifically, the high-level features are first upsampled to match the spatial resolution of the low-level features. A 1×1 convolution is then applied to align the channel dimensions. The processed high-level and low-level features are fused by elementwise addition, forming an aggregated representation that captures both detailed texture and semantic cues.

To adaptively balance the contribution from each branch, a channelwise attention mechanism is applied to the aggregated features. This mechanism computes global descriptors through average pooling and max pooling, followed by a multilayer perceptron (MLP) and a sigmoid activation to generate attention weights. Finally, the output is computed as a weighted combination of the low-level and high-level branches, where the attention weights dynamically emphasize informative channels from either source. This process can be formulated as follows:

$$F'_H = \phi\left(\mathrm{BN}\left(W_H \cdot \mathrm{Up}\left(F_H\right)\right)\right) \tag{1}$$

$$A = \sigma\left(\mathrm{MLP} * \left(\left[\mathrm{AVG}\left(F_L + F'_H\right), \mathrm{MAX}\left(F_L + F'_H\right)\right]\right)\right) \tag{2}$$

$$F_{\mathrm{out}} = A \cdot F_L + (1 - A) \cdot F'_H \tag{3}$$

where $\phi(\cdot)$ denotes the ReLU activation function, F_L and F_H represent the low-level feature map and the high-level feature, respectively, and σ is the sigmoid activation function. Specifically, regions with strong farmland characteristics are enhanced by emphasizing the high-level response, while spatially detailed areas are preserved through the low-level feature branch. The channelwise attention mechanism plays a critical role by modulating each channel's importance based on the joint feature representation, thereby refining the integration process. As a result, the final output effectively captures both semantic relevance and geometric details.

2) Differential Fusion Module: The structure of DFM is illustrated in Fig. 1(b), which is designed to enhance semantic understanding and spatiotemporal difference modeling between bi-temporal features in a CD framework. Given two input feature maps X_1 and X_2, the module performs dual-path enhancement to emphasize meaningful changes. The processing begins by computing the absolute difference and concatenation of the two feature maps, followed by convolutional transformation and refinement through a multidirectional attention module (MAM). The overall procedure can be formulated as follows:

$$F_d = |X_1 - X_2| \tag{4}$$

$$F_c = X_1 \oplus X_2 \tag{5}$$

$$Z_c = \phi\left(\mathrm{BN}_2\left(W_1 * \phi\left(BN_1\left(W_3 * F_c\right)\right)\right)\right) \tag{6}$$

$$Z_d = F_d \odot A_w \odot A_h \tag{7}$$

$$F_{\mathrm{out}} = Z_c + Z_d \tag{8}$$

where $\oplus, *$, and $\odot$ denote the channelwise concatenation, convolution, and elementwise multiplication, respectively. $\phi(\cdot)$ denotes the ReLU activation function, BN_1 and BN_2 represent the batch normalization layers, and W_3 and W_1 are the convolution kernels with kernel sizes 3×3 and 1×1, respectively. Coordinate attention generates A_w and A_h from F_d, and the resulting F_{out} enhances sensitivity to subtle changes in complex farmland areas. This dual enhancement strategy improves the network's sensitivity to subtle and fragmented changes, especially in spectrally ambiguous or morphologically complex farmland regions.

C. Farmland Soft–Hard Gate Module

After obtaining the enhanced difference features and farmland-related semantic features, these features are fed into a gating controller designed to guide the model's attention toward meaningful farmland changes. Due to the uncertainty and semantic fuzziness inherent in farmland segmentation results (particularly under limited annotation and broken-up spatial distributions), the semantic features obtained from the farmland extraction subnetwork are usually in the form of probabilistic maps rather than binary masks. This makes them unsuitable for direct hard constraints in the CD stage.

To address this, a dual-gate mechanism is introduced, as illustrated in Fig. 3. It consists of soft adaptive gate (SAG), which captures continuous spatial importance from the probabilistic farmland mask, and hybrid extractor (HE), which performs binary suppression of nonfarmland regions based on refined segmentation.

The SAG takes the predicted farmland probability map and generates a pixelwise attention map through a lightweight

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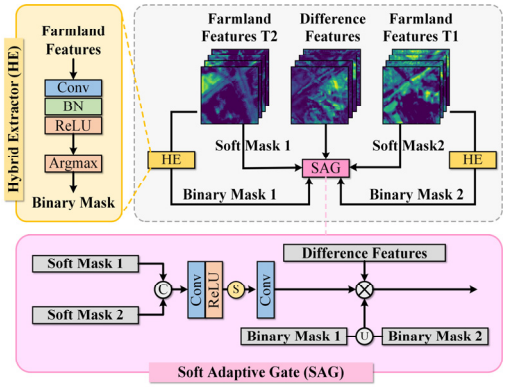


Fig. 3. FSGM.

convolutional network. This attention map is then used to scale the input feature activations, enhancing semantic regions probability for farmland. In parallel, the binary gate leverages a postprocessed binary farmland mask to suppress activations in regions that are confidently nonfarmland. These two control paths are then combined to produce a gated feature, allowing the network to focus both flexibly and selectively on farmland-related areas.

Mathematically, let F^i denote the input feature map, P_s denote the predicted soft farmland probabilistic map, and P_b denote the refined binary farmland mask. These are resized to match the input feature resolution. The gating mechanism can then be expressed as follows:

$$P_s{}' = \text{Interpolate}\,(P_s, H, W) \tag{9}$$

$$P_b{}' = \text{Interpolate}\,(P_b, H, W) \tag{10}$$

$$A_s = \sigma\left(\phi\left(\delta\left(\psi\left(P_s'\right)\right)\right)\right) \tag{11}$$

$$F_g = F^i \odot A_s \odot P_b' \tag{12}$$

$$\hat{F} = \begin{cases} F_g + F, & \text{if residual is used} \\ F_g, & \text{otherwise} \end{cases} \tag{13}$$

where ψ, δ, and ϕ represent the successive 1×1 convolution, ReLU activation, and another 1×1 convolution, respectively, used in the soft attention path; σ is the sigmoid activation function.

To refine the binary farmland mask P_b, a region-based filtering strategy is adopted. The predicted change map is first binarized using thresholding methods, such as Otsu or triangle thresholding. Then, connected region analysis is applied to identify contiguous foreground blobs. For each region, the overlap with the union of farmland masks from both time points is calculated. Only regions with overlap ratios exceeding a predefined threshold τ are retained, which can be expressed as

$$\text{Keep}\,(R_i) \Leftrightarrow |R_i \cap (M_1 \cup M_2)|/|R_i| \geq \tau \tag{14}$$

where R_i denotes the ith connected component in the change map; M_1 and M_2 are the binary farmland masks for time points T_1 and T_2, respectively; $|\cdot|$ denotes the number of pixels in a region; $\tau \in [0, 1]$ is the overlap threshold, typically set to 0.3. This dual-gate design ensures that farmland-relevant features are highlighted both softly and strictly, enabling the network to concentrate on complex farmland distributions while effectively suppressing spurious changes in irrelevant areas.

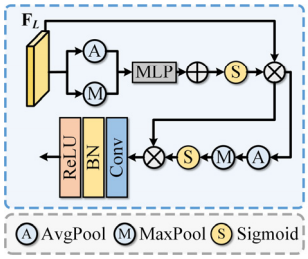


Fig. 4. MFRM.

D. Multiscale Farmland Refinement Module

To further refine change boundaries, an MFRM is introduced (Fig. 4). Although the preceding modules enable the network to capture reliable change probability maps, discrepancies still remain when compared with the ground-truth labels. Notably, most misclassifications are concentrated along the boundary regions of farmland changes, which motivates the design of MFRM for boundary optimization.

Specifically, the input feature map is first recalibrated by a channelwise attention mechanism. Global descriptors are extracted by combining average pooling and max pooling, which are then passed through an MLP and a sigmoid activation to generate attention weights. These weights are applied to rescale the original features, yielding an enhanced representation. The refined features are subsequently processed through nonlinear activation, upsampling, and batch normalization to highlight boundary-related structures. This process can be formulated as follows:

$$F_L' = \sigma\left(\text{MLP} * \left([\text{AVG}\,(F_L) + \text{MAX}\,(F_L)]\right)\right) * F_L \tag{15}$$

$$F_{\text{out}} = \text{BN}\left(\text{Up}\left(\sigma\left(\text{MAX}\left(\text{AVG}\left(F_L'\right)\right)\right)\right)\right) \tag{16}$$

where F_L denotes the input feature map, AVG and MAX represent the global average pooling and global max pooling, respectively, MLP refers to a multilayer perceptron, and σ is the sigmoid activation. By emphasizing informative channels and enhancing spatial consistency near boundaries, MFRM effectively reduces edge-related misclassifications and improves the accuracy of farmland change delineation.

E. Multitask Loss Functions

To optimize the proposed farmland CD framework, we design a multibranch loss function composed of semantic segmentation loss for farmland extraction and binary CD loss, following a phased training strategy. For farmland segmentation, a hybrid loss function combining binary cross-entropy (BCE) loss and dice loss is employed to simultaneously improve pixel-level classification accuracy and the overlap quality of the predicted regions. The two loss functions are defined as follows:

$$L_{\text{BCE}}\,(y, \hat{y}) = -\left(y \log\,(\hat{y}) + (1 - y) \log\,(1 - \hat{y})\right) \tag{17}$$

$$L_{\text{Dice}}\ \ (y, \hat{y}) = 1 - (2y\hat{y} + 1)\,/\,(y + \hat{y} + 1) \tag{18}$$

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where $y \in [0, 1]$ is the ground-truth label map and the predicted value in $\hat{y}$ lies within the range [0, 1], representing the predicted probability of the pixel being classified as farmland. During training, farmland masks are extracted from bi-temporal images, and the loss is computed independently for each timestamp. Let P denote the prediction and $y \in \{0, 1\}$ the corresponding ground truth label. For each temporal instance $t \in \{1, 2\}$, the farmland segmentation loss is defined as

$$L_{\text{farm},t} = \lambda_b \cdot \text{BCE}(P_t, y) + (1 - \lambda_b) \cdot \text{Dice}(P_t, y) \tag{19}$$

where $\lambda_b \in [0, 1]$ is a balancing coefficient used to control the relative contributions of the BCE and dice components.

The BCE loss evaluates the pixelwise classification accuracy, while the dice loss measures the spatial overlap between the predicted and ground truth regions. In this study, λ_b is empirically set to 0.75 by default.

In the CD task, the weighted BCE (WBCE) loss is adopted to address the class imbalance between changed and unchanged regions. Let $y_c \in [0, 1]$ denote the ground truth label for change regions, and $\hat{y}_c \in [0, 1]$ represent the predicted change probability. The WBCE loss is defined as follows:

$$L_{\text{cd}}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} + (1 - W) \left(\begin{matrix} W \cdot y_{c,i} \log(\hat{y}_{c,i}) \\ (1 - y_{c,i}) \log(1 - \hat{y}_{c,i}) \end{matrix} \right). \tag{20}$$

where $W \in [0, 1]$ is a weighting factor used to balance the loss contributions between changed and unchanged classes. Following previous studies, W is set to 0.25.

Finally, the total loss function is formulated by integrating the losses from both tasks (farmland segmentation and CD) and is defined as follows:

$$L_{\text{total}} = \lambda_s \cdot \left(L_{\text{farm},1} + L_{\text{farm},2} \right) + \lambda_c \cdot L_{\text{cd}} \tag{21}$$

where λ_s and λ_c are the weighting coefficients used to balance the farmland segmentation loss and the CD loss, respectively. In this study, the default values are set to 0.5 and 1.0.

III. Experiments and Results

This section presents the experiments and results, organized into six parts. In Section III-A, we introduce the two publicly available CD datasets used in our experiments. Sections III-B and III-C detail the experimental settings and evaluation metrics. In Sections III-D–III-F, we compare the proposed method with seven state-of-the-art (SOTA) approaches, evaluating performance from both quantitative and qualitative perspectives.

A. Datasets

1) CLCD Dataset [36]: The Cropland CD (CLCD) dataset contains 600 bi-temporal image pairs annotated for farmland CD, with 360, 120, and 120 pairs allocated for training, validation, and testing, respectively. The images were captured in Guangdong, China, during 2017 and 2019, by the Gaofen-2 satellite, with spatial resolutions ranging from 0.5 to 2 m. Each sample consists of two 512 × 512 bi-temporal images and a corresponding binary change mask.

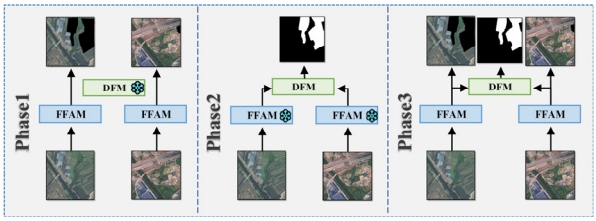


Fig. 5. Three-phase training strategy.

2) CropSCD Dataset [35]: This dataset contains 4141 pairs of 512 × 512 high-resolution images (0.5–2 m), randomly collected from rural areas in Guangdong Province, China. It covers eight types of land cover changes: water, forest, plantation, grassland, impervious surface, greenhouse, road, and bare soil. Since the proposed method is specifically designed for farmland CD, we did not perform semantic CD on the cropland semantic CD (CropSCD) dataset. Instead, we only conducted binary farmland CD.

B. Experimental Details

All experiments were conducted on an NVIDIA Quadro RTX 5000 GPU with 16 GB of memory, using CUDA 11.7 and PyTorch 1.13.1. Model training was carried out separately on the CLCD and CropSCD datasets. To ensure fair comparisons, the training settings of seven existing comparison models followed the recommended configurations provided in their original studies. Due to GPU memory limitations, the original 512 × 512 size images in the CLCD and CropSCD datasets were cropped into 256 × 256 patches for training. Each model was trained for 200 epochs to ensure convergence, with a batch size of 8.

To improve the performance of the farmland CD model, a staged training strategy was adopted. As illustrated in Fig. 5, the training process was divided into three phases, each targeting different components of the network.

In Phase 1 (epochs 0–49), the farmland segmentation branches were pretrained, while all CD modules were frozen, enabling the model to learn reliable farmland representations under the supervision of the segmentation loss. In Phase 2 (epochs 50–99), the segmentation branches were fixed and only the CD branch was optimized, allowing the model to focus on learning discriminative CD features guided by the CD loss. In Phase 3 (epochs ≥ 100), all parameters were jointly fine-tuned in an end-to-end manner to further integrate complementary information from both farmland extraction and CD modules. To enhance training stability and reduce overfitting, a smaller learning rate was adopted in the final stage. During this phase, the total loss is employed to impose constraints on the task. Through the three-phase training process, the model not only achieves higher accuracy but also improves computational efficiency, reducing the overall training cost. Experimental statistics show that the number of parameters updated in each phase gradually increases: 32.46 M in Phase 1, 39.91 M in Phase 2, and 48.02 M in Phase 3.

In the farmland feature extraction stage, using CD labels as supervision would lead to an illogical training paradigm,

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as the objective of farmland extraction is inherently different from that of CD. Therefore, farmland detection labels must be used directly as supervision. However, obtaining such labels poses a challenge. To address this issue, this article adopts a weakly supervised farmland extraction strategy to generate the required labels. In the first generation of farmland labels, the annotations were created independently. Specifically, this article randomly selected 5% of the training samples and manually annotated the farmland regions. These annotated samples were then used to train a farmland extraction network, which subsequently predicted farmland masks for the entire dataset. The predicted results were stored as the next generation of farmland labels and served as supervision for the first phase of our training pipeline.

Notably, highly accurate farmland masks are not required during training, as they mainly serve to guide the CD task; thus, coarse masks covering most farmland regions are sufficient and reduce annotation cost. The AdamW optimizer ($\beta 1 = 0.9$, $\beta 2 = 0.999$, and weight_decay $= 0.01$) is used throughout training. Across different training phases, the optimizer and learning rate scheduler are reconfigured to update only the relevant branch parameters.

C. Evaluation Metrics

To evaluate the performance of the models in our experiments, we adopted five classical evaluation metrics: precision, recall, $F1$, intersection over union (IoU), and overall accuracy (OA). Precision evaluates the correctness of the model's positive predictions, while recall measures its ability to identify all actual positive samples. $F1$ is the harmonic mean of precision and recall. IoU quantifies the overlap between the predicted change regions and the ground truth. OA reflects the overall correctness by computing the ratio of correctly predicted instances to the total number of instances. These metrics are computed as follows:

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{22}$$

$$\text{Rec} = \text{TP}/(\text{TP} + \text{FN}) \tag{23}$$

$$F1 = 2 \times \text{Rec} \times \text{Pre}/(\text{Rec} + \text{Pre}) \tag{24}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{25}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}). \tag{26}$$

These evaluation metrics are calculated based on the confusion matrix, which consists of true positives (TPs), FPs, true negatives (TNs), and FNs.

D. SOTA Methods for Comparison

To verify the advanced performance of the proposed method, seven SOTA CD methods were selected for comparative experiments. They are as follows.

1) FC-EF [17]: It is built on the UNet architecture, where the two input images are concatenated along the channel dimension and treated as multispectral inputs. Feature fusion is then achieved through skip connections, linking the encoder with the corresponding decoder layers at each scale.
2) ChangeFormer [41]: It is a transformer-based RSCD approach that leverages the powerful modeling capabilities of the vision transformer (ViT). By introducing a feature-enhanced fusion strategy and a multiscale modeling scheme, it effectively addresses the limitations of conventional CNNs in capturing long-range dependencies.
3) MSCANet [35]: It adopts a CNN–transformer architecture with multiscale context aggregation for farmland CD. CNNs are used to extract hierarchical features, while a transformer-based multiscale attention module encodes and aggregates contextual information. A multibranch CNN head then produces the CD map under deep supervision.
4) A2Net [32]: It introduces a novel lightweight network that detects changes by leveraging features extracted from mobile networks, enhanced through progressive feature aggregation and supervised attention.
5) HGINet [42]: It is a multitask network for semantic CD in high-resolution RS images, combining segmentation and CD to better model temporal correlations between bi-temporal features and their differences, improving accuracy in complex scenarios.
6) MeGNet [36]: It is a memory-guided network tailored for semantic farmland CD, featuring a class-aware memory module that captures and differentiates semantic representations of various change categories, mitigating the challenge posed by diverse change types and limited annotated samples.
7) FEFNet [43]: It is a feature enhancement and feedback network for RSCD, designed to balance accuracy and efficiency. It introduces a dual-feature fusion enhancement module to strengthen feature representation between bi-temporal images and a feature coupling feedback module to decode multiscale change features with global awareness.

E. Quantitative Evaluation

The evaluation results of the proposed method and other comparison methods on the CLCD dataset are shown in Table I, indicating that FarmCD achieves superior performance over all comparison methods in terms of accuracy-related metrics. Specifically, compared to FC-EF, ChangeFormer, MSCANet, A2Net, HGINet, MeGNet, and FEFNet, FarmCD achieves higher $F1$ with improvements of 17.06%, 6.42%, 3.17%, 29.14%, 10.17%, 7.45%, and 8.32%, respectively. It also attains higher IoU, with increases of 18.82%, 7.65%, 3.87%, 29.65%, 11.78%, 8.8%, and 9.77%, respectively.

Similarly, on the CropSCD dataset, FarmCD achieves SOTA performance, as shown in Table I. Specifically, compared to FC-EF, ChangeFormer, MSCANet, A2Net, HGINet, MeGNet, and FEFNet, FarmCD demonstrates $F1$ improvements of 5.36%, 4.19%, 4.14%, 10.35%, 6.27%, 7.04%, and 5.24%, respectively. Concurrently, it also attains higher IoU, with relative gains of 5.79%, 4.57%, 4.51%, 10.8%, 6.74%, 7.52%, and 5.67%.

The superior performance of FarmCD is mainly attributed to its three-stage training strategy. In the first stage, the network

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TABLE I
QUANTITATIVE COMPARISONS ON CD RESULTS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods		FC-EF	ChangeFormer	MSCANet	A2Net	HGINet	MeGNet	FEFNet	FarmCD
Year		2018	2022	2022	2023	2024	2024	2025	ours
FLOPs(G)		3.58	21.18	14.8	3.05	12.65	22.52	11.24	15.95
Params(M)		1.35	29.75	16.42	3.78	27.7	41.07	2.55	48.02
CLCD (%)	OA	94.48	95.21	95.69	91.99	94.76	95.3	94.86	**96.19**
	IOU	39.44	50.61	54.39	28.61	46.48	49.46	48.49	**58.26**
	F1	56.57	67.21	70.46	44.49	63.46	66.18	65.31	**73.63**
	Pre	68.28	68.55	71.93	45.98	65.97	71.31	65.61	**78.54**
	Rec	48.29	65.91	69.04	43.1	61.15	61.76	65.03	**69.31**
CropSCD (%)	OA	94.72	96.72	96.68	96.33	94.76	96.57	96.69	**96.94**
	IOU	44.22	45.44	45.5	39.21	43.27	42.49	44.34	**50.01**
	F1	61.32	62.49	62.54	56.33	60.41	59.64	61.44	**66.68**
	Pre	70.06	72.66	72.12	73.03	70.36	73.08	74.27	**75.82**
	Rec	54.53	54.81	55.21	45.85	52.92	50.38	52.38	**59.51**

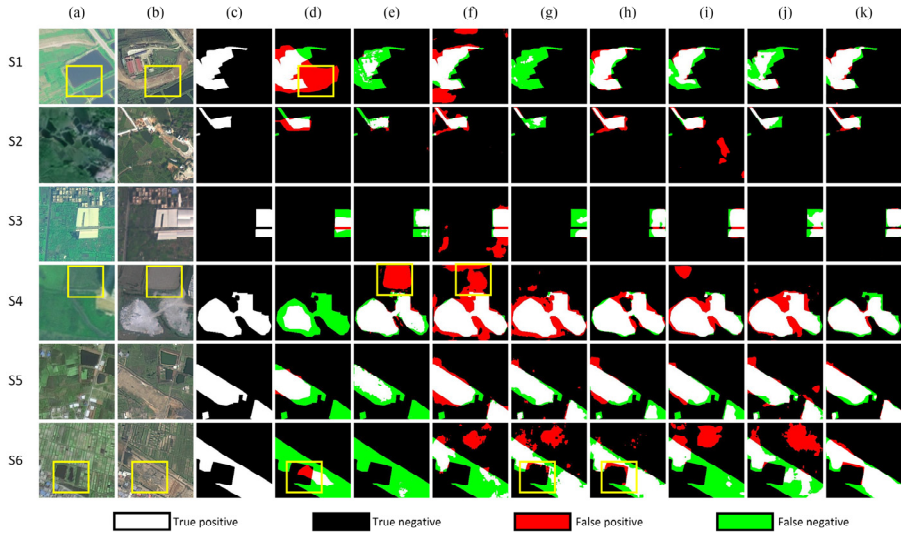


Fig. 6. Visual comparisons of different methods on the CLCD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) A2Net. (e) FC_EF. (f) ChangeFormer. (g) HGINet. (h) MSCA. (i) MeGNet. (j) FEFNet. (k) Proposed FarmCD. S1–S6 represent six representative samples.

is initialized using cropland guidance to learn preliminary representations. The second stage leverages these pretrained parameters to focus on CD. In the final joint fine-tuning stage, a dual-gate mechanism with soft and hard constraints incorporates cropland information to emphasize relevant cropland changes while suppressing irrelevant interference. This systematic training scheme effectively explains the strong performance of FarmCD.

F. Qualitative Evaluation

The visual comparison of different methods across the two datasets is presented in Figs. 6 and 7, where distinct colors are TP (white), FPs (red), TNs (black), and FNs (green). Our method demonstrates superior performance in the following aspects.

1) Superior Edge and Detail Preservation: A comprehensive evaluation of the detection results demonstrates that the proposed FarmCD exhibits clear advantages in capturing fine-grained changes and detailed cropland boundaries. This superior performance is fundamentally attributed to its specialized design tailored to the unique characteristics of farmland CD. Farmland CD differs markedly from conventional CD tasks. Cropland areas typically exhibit fragmented spatial distributions and irregular geometries, in contrast to the relatively regular and structured patterns of urban buildings. As a result, traditional CD networks particularly those designed for building detection often struggle to handle such irregular cropland patterns. FarmCD effectively overcomes these limitations through several key innovations. By incorporating cropland-specific mechanisms, the network enhances its sensitivity to agricultural features, enabling more accurate extraction of intricate edge details characteristic of farmland landscapes. These design choices collectively lead to substantially improved visual quality and reliability of the CD results.

Experimental results further confirm that FarmCD consistently detects fragmented cropland boundaries and subtle changes that are frequently missed or inaccurately identified by other SOTA methods. Capturing such fine-grained details is critical for precise agricultural change analysis. While conventional networks tend to fail in these challenging scenarios, FarmCD maintains robust performance even

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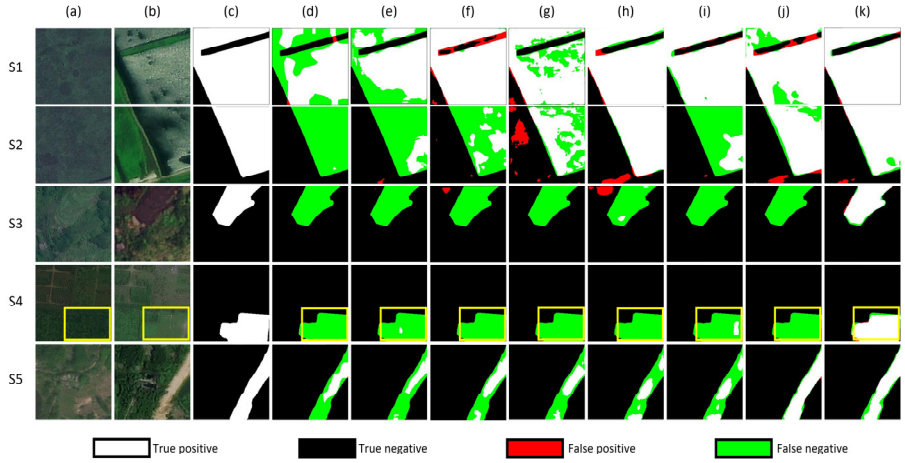


Fig. 7. Visual comparisons of different methods on the CropSCD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) A2Net. (e) FC_EF. (f) ChangeFormer. (g) HGINet. (h) MSCA. (i) MeGNet. (k) Proposed FarmCD. S1–S5 represent five representative samples.

under complex conditions involving irregular and fragmented cropland boundaries. These results highlight the importance of domain-specific network design for effective farmland CD.

2) Cross-Style Image Accurate Detection Capability: Due to significant variations in image styles, including illumination, and imaging conditions across regions and time periods, as shown by S1–S6 in Fig. 6, the CLCD dataset poses a major challenge for CD models. Under such conditions, most existing methods suffer from extensive FNs or FPs, revealing their limited generalization capability.

In contrast, our method exhibits stronger robustness and generalization. As shown in the visual results, even under substantial style differences between image pairs, the model effectively suppresses false alarms and reduces missed detections. This suggests that the network captures more domain-invariant and discriminative change features, enabling accurate localization of true change regions. We attribute this improvement to the design of FFAM and DFM, which enhance the model's ability to aggregate consistent semantic representations across domains.

3) Robust Anti-Interference Capability: Another critical challenge in farmland CD lies in the frequent misclassification of water bodies or forested areas as cropland changes. Although these land cover changes are irrelevant to actual cropland monitoring tasks, their spectral similarities to croplands in RS imagery can significantly interfere with detection accuracy, ultimately compromising the reliability of CD results. As clearly illustrated by the yellow boxes in sample S6 in Fig. 6 and sample S4 in Fig. 7, this interference manifests in two distinct but equally problematic scenarios.

In Fig. 6, the yellow boxes highlight the regions affected by interference from water bodies. Pond variations can degrade detection performance in two ways: direct misclassification, where water changes are mistaken for cropland changes, and spatial propagation, where the effect spreads to adjacent areas. During training, the network may overgeneralize the invariant labeling of ponds, suppressing CD in neighboring cropland and causing extensive FNs. FarmCD mitigates this issue by maintaining cropland-specific sensitivity, accurately handling water bodies while detecting genuine changes in surrounding farmland, thus enhancing interference resistance.

Fig. 7 (S4 example) illustrates another challenge: spectral confusion between croplands and forests, indicated by the yellow boxes. Conventional CD networks often fail to distinguish these classes due to similar spectral signatures, leading to large omission errors. For instance, when cropland converts to forest, the network may misclassify both periods as the same land cover, missing the change entirely. FarmCD overcomes this by capturing subtle discriminative features between spectrally similar classes. As shown in the results, it successfully detects the full transition area, whereas comparison methods fail.

These two cases jointly demonstrate FarmCD's superior ability to maintain detection accuracy under strong interference. This performance stems from three key factors: 1) enhanced discrimination of cropland-specific features; 2) context-aware modeling of spatial relationships among different land-cover types; and 3) robust feature representations that preserve sensitivity to genuine changes while suppressing irrelevant variations. Together, these capabilities enable FarmCD to achieve high precision in complex real-world scenarios.

IV. Discussion

In this section, we conducted a series of experiments on the network and obtained several interesting findings.

A. Training Strategy Ablation Study

To evaluate the impact of different training strategies on farmland CD performance, four schemes were compared.

1) *Ours*: A staged training strategy where farmland segmentation is trained first, followed by CD training, and finally joint fine-tuning.

2) *E2E (End-to-End)*: Training all parameters simultaneously without stage division, with both tasks initialized from scratch.

3) *CD-Only*: Training only the CD stage while freezing the farmland segmentation stage, to verify its contribution.

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TABLE II
ABLATION STUDY ON TRAINING STRATEGIES

Method	S_1	S_2	S_3	OA (%)	IoU (%)	F1 (%)	Pre (%)	Rec (%)
Ours	✓	✓	✓	**96.19**	**58.26**	**73.63**	**78.54**	69.31
E2E	×	×	✓	96.13	57.94	73.38	77.58	**69.59**
CD-only	×	✓	×	95.75	53.19	69.45	77.44	62.95
W/o FT	✓	✓	×	95.83	55.71	71.56	75.08	68.35

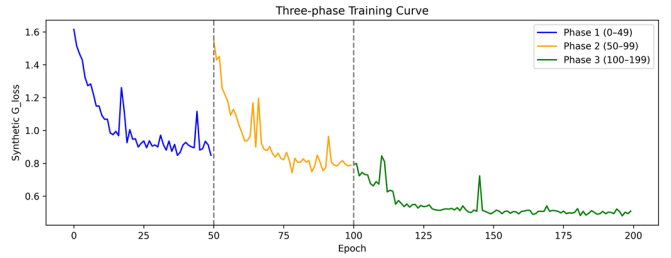


Fig. 8. Three-phase training curve.

4) *W/o Fine-Tune (FT)*: Applying stagewise training without the final joint fine-tuning stage.

The evaluation results are presented in Table II, where S_x indicates whether the xth training stage is included, measured by OA, IoU, $F1$, precision, and recall. The three-phase training strategy achieves the highest OA (96.19%), IoU (58.26%), $F1$ (73.63%), and precision (78.54%), indicating superior performance compared to the other strategies.

Compared with the end-to-end training approach, the three-phase strategy shows improved IoU and precision, while maintaining comparable recall, suggesting that progressive optimization enhances the model's ability to detect changes accurately. Both CD-only and W/o FT strategies exhibit reduced performance, especially in IoU and $F1$, which confirms the importance of farmland segmentation supervision and joint fine-tuning in improving CD accuracy and reliability. Moreover, the loss curves of the three-phase training strategy, as illustrated in Fig. 8, further validate the effectiveness of our design. The curve exhibits a clear progressive optimization pattern that aligns well with the intended behavior of each training stage, demonstrating the rationality and stability of the proposed multistage optimization scheme.

B. Module Ablation Experiments

1) FFAM and DFM: The proposed multiscale farmland-aware feature extractor serves as the core of the network by integrating farmland semantic priors with difference-enhanced features. To assess its effectiveness, ablation studies were conducted on its two submodules: FFAM and DFM. Removing FFAM ($\text{FarmCD}_{\text{noFFAM}}$) eliminated farmland guidance, while removing DFM ($\text{FarmCD}_{\text{noDFM}}$) replaced learnable fusion with naive subtraction. Eliminating both resulted in $\text{FarmCD}_{\text{noFFAMandDFM}}$.

From the results shown in Table III, the removal of either submodule resulted in a notable performance decline. Specifically, removing FFAM led to a 2.63% IoU drop and a 2.14% $F1$ decrease. The absence of DFM caused a 3.06% drop in IoU. When both modules were removed, IoU dropped by 6.12%, showing the compound effect. These results indicate that FFAM helps the model focus more effectively on farmland-related features, while DFM improves the precision of change localization. Together, they form a powerful combination for targeted CD. This confirms the effectiveness of farmland-aware attention and differential enhancement as mutually beneficial components in the network design.

TABLE III
ABLATION STUDY RESULTS

No.	Variants	IoU (%)	F1 (%)	Rec (%)
#01	FarmCD	58.26	73.63	69.31
#02	$\text{FarmCD}_{\text{noFFAM}}$	55.63	71.49	64.97
#03	$\text{FarmCD}_{\text{noDFM}}$	55.20	71.13	66.79
#04	$\text{FarmCD}_{\text{noFFAMandDFM}}$	52.14	68.54	64.06
#05	$\text{FarmCD}_{\text{noFSGM}}$	53.86	70.01	66.34
#06	$\text{FarmCD}_{\text{noMFRM}}$	56.79	72.44	66.70

2) Farmland Soft–Hard Gate Module: To evaluate the effect of farmland region-aware gating, we ablated FSGM, which integrates soft attention (from probability maps) and hard suppression (from binary masks). In $\text{FarmCD}_{\text{noFSGM}}$, both gates were removed, leaving difference features unconstrained. As shown in Table III, this led to a 4.4% IoU and 3.62% $F1$ drop, along with reduced recall, confirming the necessity of explicit regional guidance. The results demonstrate that combining soft and hard masks enhances focus on farmland areas, improving localization and reducing false alarms.

3) Multiscale Farmland Refinement Module: To assess the role of MFRM, we removed it to form $\text{FarmCD}_{\text{noMFRM}}$. Without MFRM, the model lost the ability to refine features with hierarchical farmland priors. As shown in Table III, IoU and $F1$ dropped by 1.47% and 1.19%, respectively. Although the decline is smaller than that from removing core modules (e.g., FSGM, FFAM, and DFM), the results confirm MFRM's supportive role in reinforcing feature consistency and boundary precision in heterogeneous farmland scenes.

C. Analysis of Change Categories

In the experiments conducted on the CropSCD dataset, farmland change categories were divided into eight types: water, forest, plantation, grassland, impervious surface, greenhouse, road, and bare soil. In the labels, all changes are defined as interactions between farmland and these eight land-cover classes. To investigate the impact of change categories on farmland CD accuracy, the precision of changes between farmland and each class was calculated separately.

Fig. 9 illustrates the distribution of misdetections (FP) and ground truth pixels (GT) across the eight categories. The bar chart on the upper left shows the proportion of FP contributed by each category relative to the total FP, alongside the proportion of GT area contributed by each category relative to the total GT. The bar chart on the lower left illustrates the FP-to-GT ratio for each category, where a lower ratio indicates higher detection accuracy. The bubble chart on the right illustrates

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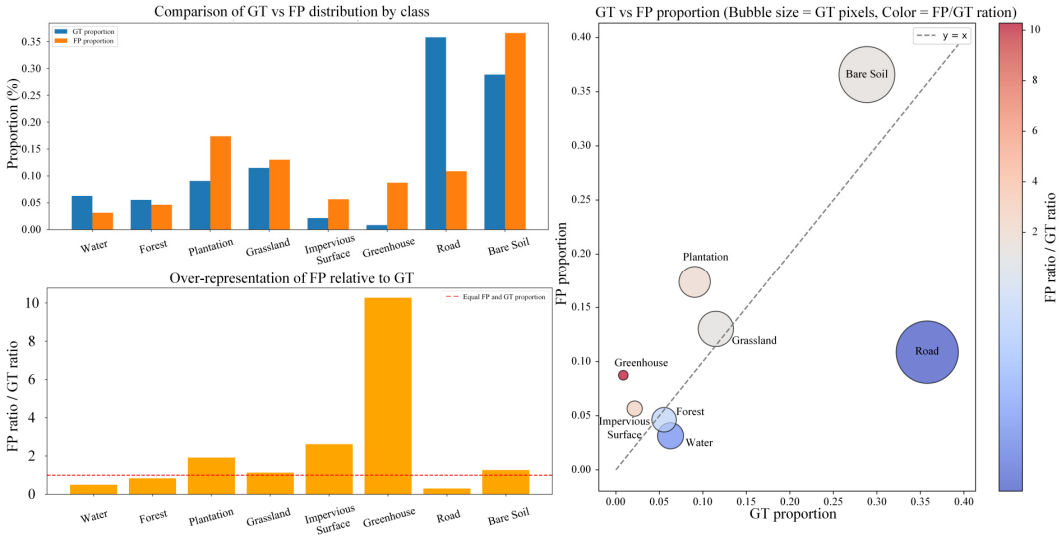


Fig. 9. Proportion of each category and the situation of false detections.

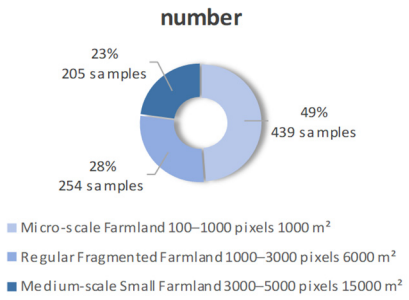


Fig. 10. Tree type of farmland.

the detection performance across different categories. The circle area represents the proportion of changes for each category, while the color denotes the FP-to-GT ratio, with red indicating higher misdetection rates. The results show that bare soil and greenhouse contribute most to detection errors. Bare oil accounts for a large proportion of total changes, whereas greenhouse, despite representing only about 1% of all changes, contributes disproportionately to misdetections, reaching nearly 9%. This is mainly due to the limited number of greenhouse samples in the training set, which hinders effective learning of this class. In contrast, categories with relatively sufficient training samples, such as water and forest, exhibit fewer misdetections, despite remaining challenging in farmland CD tasks.

By combining the observations from Fig. 9, it can be seen that land-cover types, such as water (mainly ponds), forest, and grassland, which are typically difficult to distinguish from farmland, achieve relatively favorable detection results when interacting with farmland changes. This finding clearly demonstrates the effectiveness of the feature extraction modules in the proposed network.

D. Regarding Fragmented Small Farmland

To further verify the effectiveness of the proposed method in detecting internal changes within fragmented farmland areas, we conducted targeted experiments on the regions with different farmland parcel size characteristics. In Fig. 10, the input images used in the experiments cover three fragmentation categories defined by pixel area.

For the data that meet these three category conditions, we extract them for experiments and evaluate the results using four metrics: IoU, $F1$, precision, and recall. The comparative results shown in Fig. 11 (IoU, F1, precision, and recall under different farmland fragmentation conditions):

1) *For Microscale Farmland (100–1000 pixels)*: FarmCD achieves an IoU of 12.07%, an $F1$ of 21.53%, a precision of 23.87%, and a recall of 19.61%. These results outperform all other comparison methods, demonstrating better sensitivity to fine-grained and fragmented change areas.
2) For *Regular Fragmented Farmland (1000–3000 pixels): F*armCD achieves an $F1$ of 36.22% and IoU of 22.11%, surpassing other methods. This indicates that the method maintains robust performance in commonly fragmented agricultural regions.
3) *For Medium-Scale Small Farmland (3000–10 000 pixels)*: FarmCD continues to show superior results with an $F1$ of 44.33% and IoU of 28.48%, proving its capability across larger and less fragmented change areas.

The proposed approach demonstrates consistent advantages across all fragmentation scales, especially in detecting changes in highly fragmented farmland parcels. This highlights its practical value for real-world applications, where farmland varies widely in size and shape.

E. Anti-Interference Analysis

To further assess the model's robustness to radiometric variations, controlled perturbation experiments were conducted on the validation and test sets. As illustrated in Fig. 12,

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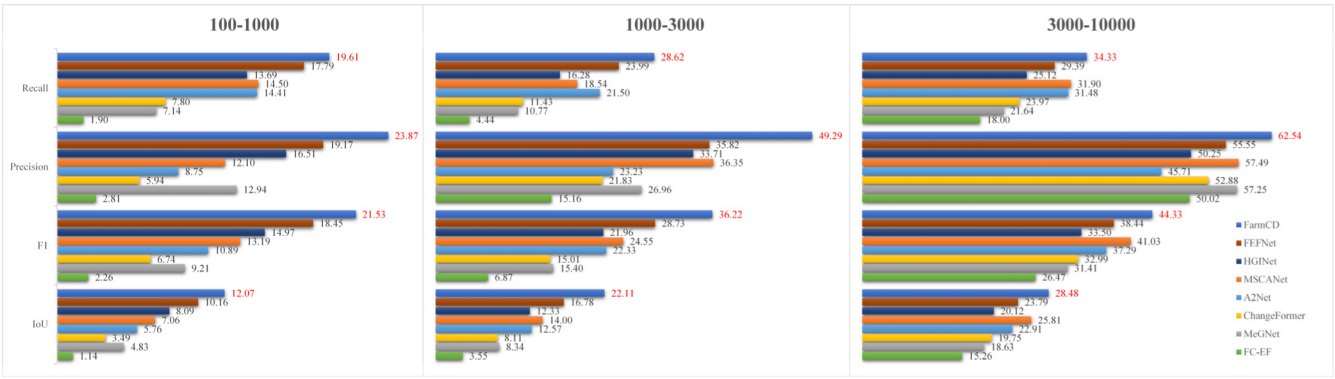


Fig. 11. Result in three fragmentation categories defined by pixel area.

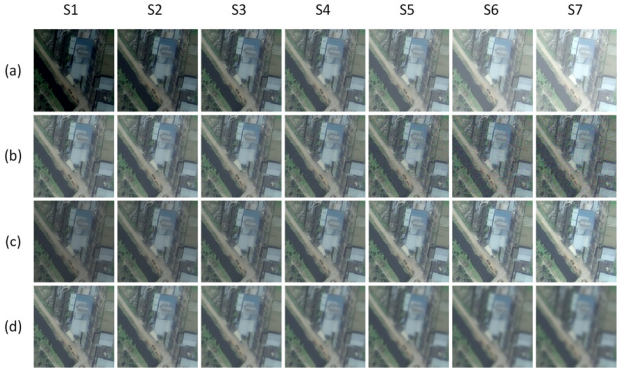


Fig. 12. Illustration of perturbed images. (a) Brightness shift ($\Delta b \in \{-0.2, -0.1,..., 0.2\}$). (b) Gaussian noise ($\sigma \in \{0.005,..., 0.05\}$). (c) Color shift ($\gamma \in \{-0.02,..., 0.02\}$). (d) Gaussian blur ($k \in \{3, 5,..., 9\}$). (e) S1–S7 represent the perturbation levels.

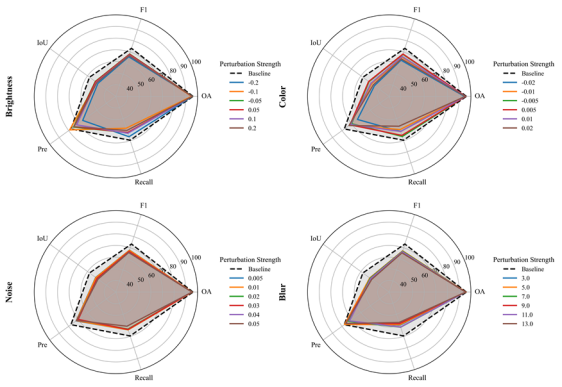


Fig. 13. Results of the anti-interference experiment.

four types of perturbations were considered: brightness shift ($\Delta b \in \{-0.2, -0.1,..., 0.2\}$), color shift ($\gamma \in \{-0.02,..., 0.02\}$), Gaussian noise ($\sigma \in \{0.005,..., 0.05\}$), and Gaussian blur ($k \in \{3, 5,..., 9\}$). For each perturbation type, multiple intensity levels were applied, and the results were compared with the baseline (no perturbation).

The outcomes are illustrated in Figs. 13 and 14. Fig. 13 presents the radar plots summarizing the overall performance changes across five evaluation metrics (OA, precision, recall, $F1$, and IoU) under different perturbation types. Fig. 14 depicts

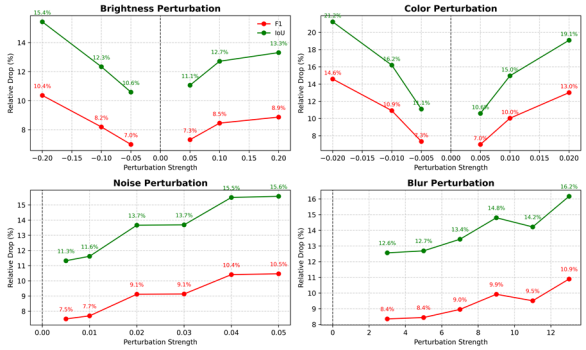


Fig. 14. Relative drop of metrics under different perturbations.

the relative performance degradation of $F1$ and IoU with increasing perturbation strength, where positive and negative perturbations (e.g., for brightness and color) are plotted separately. From these results, several observations can be made. First, the performance consistently decreases with stronger perturbations, indicating that radiometric variations have a substantial impact on CD accuracy. Second, the effects of brightness and color shifts are asymmetric: negative shifts cause more severe degradation than positive ones, and accuracy drops rapidly as the magnitude of negative perturbations increases. This suggests that the model is particularly sensitive to spectral distortions when distinguishing farmland from other land-cover types.

Fig. 13 shows that the absolute accuracy remains relatively stable within moderate perturbation ranges, with most perturbed results clustering around the baseline. This demonstrates that the proposed network maintains robustness against radiometric disturbances.

F. Feature Maps in FarmCD

To verify the effectiveness of each feature extractor, we visualized the feature maps at different stages of FarmCD. Fig. 15 presents several visualization results of the features extracted by the FarmCD feature extractor on three sample pairs. Specifically, Fig. 15(a) shows the original T1 image and the farmland features extracted by FFAM, Fig. 15(b) shows the T2 image and its corresponding farmland features extracted by FFAM, and Fig. 15(c) depicts the change regions

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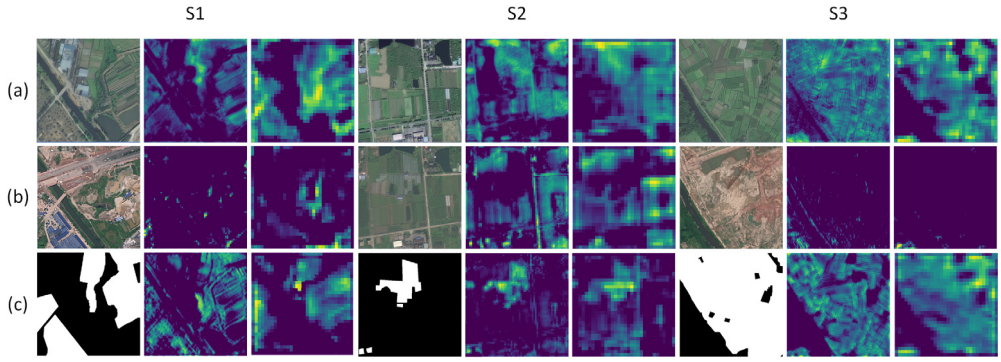


Fig. 15. Visualization results of FarmCD's features. (a) T1 image feature map from FFAM. (b) T2 image feature map from FFAM. (c) Fusion feature from FSGM. S1–S3 represent three representative samples.

and the fusion feature extracted by FSGM. The visualization results demonstrate that the feature extractors exhibit strong sensitivity to farmland areas.

V. Conclusion

Farmland CD has become increasingly important, yet traditional methods often struggle in rural regions due to spectral heterogeneity, seasonal variability, and complex field structures. To address these challenges, this research proposes FarmCD, a farmland-specific CD framework. The method employs FFAM and DFM to extract and fuse semantic farmland features and temporal differences, consisting of FFAM for adaptive multiscale alignment and DFM for enhancing discriminative change cues. In addition, FSGM integrates soft attention maps with refined binary masks derived from farmland predictions, constraining the network to focus on meaningful agricultural changes.

Extensive experiments and ablation studies demonstrate that each component contributes to performance improvements, with the complete FarmCD model achieving superior results over ablated variants, particularly in complex rural landscapes, where precise farmland delineation is crucial. Nonetheless, challenges remain in accurately capturing field boundaries, embankments, and highly fragmented or small-scale farmland changes, highlighting directions for future research.

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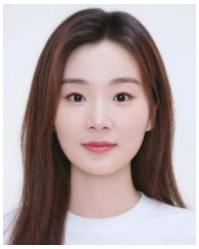
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